



Effect of the Reduction Temperature of Co-La-Zr/AC on the Synthesis of Higher Alcohols from Syngas

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Abstract: CO hydrogenation over a Co-La-Zr/AC catalyst prepared by co-impregnation and reduced at different temperatures was studied. The catalyst reduced at 703 K showed good activity at 3.0 MPa, 495 K, H_2/CO (V/V) = 2, and GHSV = 500 h^{-1} . CO conversion and selectivity for total alcohols were 59.1% and 42.0%, respectively, and C_6 – C_{18} higher alcohols were 50.4% of the alcohols. Metallic cobalt and cobalt carbide were detected by XRD as the main cobalt species. The reduction temperature affects both the cobalt dispersion and formation of cobalt carbide. Cobalt carbide may play an important role in the synthesis of higher alcohols on Co-La-Zr/AC.

Key words: cobalt catalyst; FT synthesis; reduction temperature; high alcohol; cobalt carbide

The utilization of natural gas has attracted increasing attention due to the potential shortage of petroleum reserves, and the direct synthesis of mixed alcohols from syngas (H_2 and CO) is a field of interest. Catalysts for producing mixed alcohols from syngas include modified methanol synthesis catalysts, alkali-doped molybdenum sulfide catalysts, and modified Fischer-Tropsch (FT) synthesis catalysts [1–5]. C_1 – C_6 alcohols can be successfully produced from syngas over the above catalysts. Few reports have considered the direct synthesis of C_6 – C_{18} alcohols with improved yields from syngas. These are the feed stocks for plasticizers, detergents, and lubricants. The modified FT synthesis catalysts are thought to be attractive for industrial applications due to their high activity, high selectivity toward higher alcohols, and moderate reaction conditions. Here, we report on the influence of reduction temperature on the performance of a Co-La-Zr/AC catalyst in CO hydrogenation.

The Co-La-Zr/AC catalyst was prepared by co-impregnation. Activated carbon made from coconut shell (20–40 mesh, BET surface area 893 m^2/g , and pore volume 0.57 cm^3/g) was dried for 0.5 h and impregnated in a rotary evaporator at RT with an aqueous solution of lanthanum nitrate, zirconium nitrate, and cobalt nitrate mixture. After evaporation of the water and

drying in air at 363 K for 8.0 h, the sample was calcined at 623 K for 3.0 h under argon. The loadings of Co, Zr, and La were 15%, 1.0%, and 0.5%, respectively.

Activity evaluation was carried out in a stainless steel fixed-bed microreactor. The samples (4 ml) were reduced in situ in a flow of H_2 for 12 h before reaction. Data were taken 48 h after the reaction started and a steady state activity had appeared. The product gases were analyzed online by an Agilent 3000A Micro GC and TCD detector. The liquid organic products and aqueous products were analyzed offline with a HP 6890 GC and FID detector. XRD analysis was carried out using a PANalytical X'Pert PRO diffractometer. Temperature-programmed reduction (TPR) experiments were performed on a Micromeritics Autochem 2910 apparatus, and an Omnistar 300 quadrupole mass spectrometer was used for detection.

CO hydrogenation results over the Co-La-Zr/AC catalysts are presented in Table 1. The products contained C_1 – C_{18} alcohols, C_1 – C_{21} hydrocarbons, and CO_2 . Almost all the alcohols were linear α -alcohols, which obeyed the Schulz-Flory distribution. When the reduction temperature was increased from 623 to 703 K, the CO conversion increased from 25.9% to 59.1%, and the selectivity for alcohols increased from 33.2% to 42.0%. The concentration of C_6 – C_{18} higher alcohols reached

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Table 1 Effect of reduction temperature on CO hydrogenation activity over the Co-La-Zr/AC catalyst

Reduction temperature (K)	CO conversion (%)	Selectivity (%)				Alcohol distribution (%)			α_{ROH}
		HC ₁₋₄	CO ₂	HC ₅₊	ROH	C ₁	C ₂₋₅	C ₆₋₁₈	
623	25.9	36.3	6.4	24.1	33.2	7.9	61.8	30.3	0.58
653	43.5	33.2	4.8	26.2	35.8	6.1	58.1	35.8	0.59
673	56.9	29.7	4.9	29.2	36.2	3.8	51.5	44.7	0.71
703	59.1	21.6	4.1	32.3	42.0	4.0	45.6	50.4	0.73
723	51.4	28.5	4.8	27.6	39.1	5.0	56.9	38.1	0.63

Reaction conditions: H₂/CO (*V/V*) = 2, GHSV = 500 h⁻¹, *T* = 495 K, *p* = 3.0 MPa.

α_{ROH} : Chain growth probability of alcohols.

50.4% in the total alcohols. CO conversion and alcohol selectivity decreased with the further increase of reduction temperature. We can conclude that the reduction temperature played a pivotal role in the CO hydrogenation performance of the Co-La-Zr/AC catalysts.

Fig. 1 shows the XRD patterns of the Co-La-Zr/AC catalysts reduced at different temperatures. Diffraction peaks that can be attributed to metallic cobalt (Co⁰) and CoO appeared on the samples before the FT reaction. Using the Scherrer equation, the cobalt particle sizes on the samples reduced at 623, 653, 673, 703, and 723 K were calculated as 32.1, 20.9, 14.9, 12.6,

and 14.0 nm, respectively. We conclude that the reason for the activity increase when the reduction temperature increased from 623 to 703 K was the higher cobalt dispersion.

The major products are hydrocarbons with only a small amount of oxygenates with traditional Co-based catalysts. It is well known that Co⁰ is responsible for the formation of hydrocarbons. Interestingly, the alcohol selectivity was surprisingly high at 42.0% in the case of reduction at 703 K. This implied that there may be active species other than metallic cobalt particles on the surface of the Co-La-Zr/AC catalyst. The XRD results showed the existence of cobalt carbide (Co₂C) species besides Co⁰ on the used Co-La-Zr/AC catalysts (Fig. 1(b)). The transition metal carbides usually exhibit noble metal-like characteristics and are able to adsorb CO without dissociation, and Co₂C species was reported to have an important role in the formation of alcohols [6]. Accordingly, we believe that the Co₂C species are the active sites for the insertion of adsorbed CO while Co⁰ particles are the active sites for the dissociation of adsorbed CO and growth of carbon chains on the Co-La-Zr/AC catalysts. The typical diffraction peaks due to Co₂C species were much sharper for samples reduced at lower temperature while they became broader with increased reduction temperature. This suggests that the reduction temperature affects the formation of Co₂C species and, in turn, both the activity and alcohol selectivity.

Because the reduction temperature has a strong effect on FT synthesis over the Co-La-Zr/AC catalyst, catalyst reducibility was further investigated by H₂-TPR experiments. Fig. 2 gives the H₂-TPR profiles over the Co-La-Zr/AC catalyst. The first two hydrogen consumption peaks at 480 and 595 K were due to the two-step reduction of Co₃O₄ to CoO and then to Co⁰ [7,8], while the broad peak at higher temperature was due to the hydrogenation of oxygen-containing groups on the surface of activated carbon. Strong signals for CH₄, CO, and CO₂ were detected above 600 K, which implied the existence of abundant oxygen-containing organic groups on the surface of activated carbon. We can postulate that these oxygen-containing organic groups affect the dispersion of cobalt particles. It was reported that the Pt particle size on a Pt/C catalyst decreased with an increase in reduction temperature [9], and it was suggested that the faster decomposition of oxygen-containing organic groups

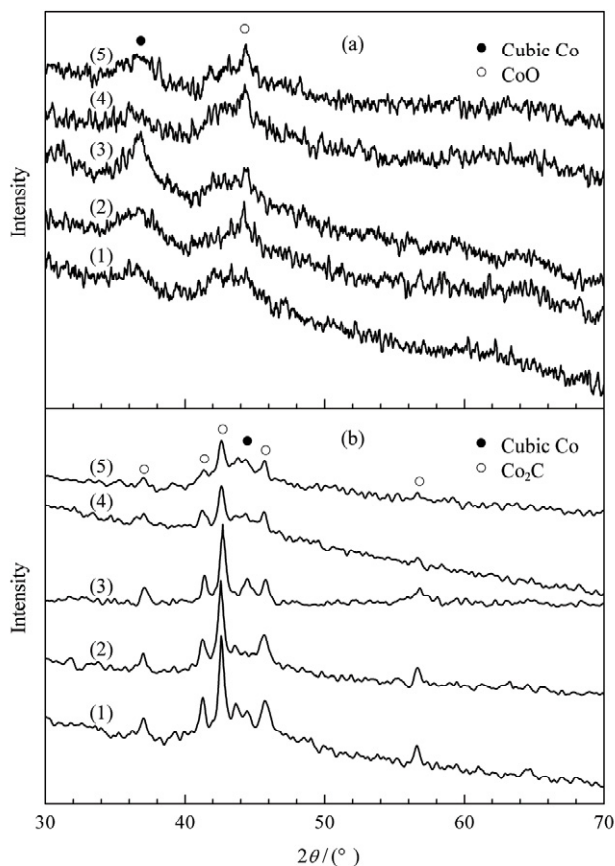


Fig. 1. XRD patterns of the Co-La-Zr/AC samples reduced at different temperatures. (a) Before reaction; (b) After reaction. Reduction temperature: (1) 623 K; (2) 653 K; (3) 673 K; (4) 703 K; (5) 723 K.

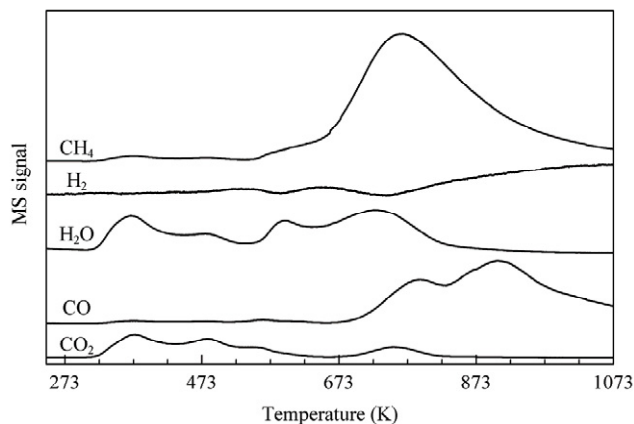


Fig. 2. H₂-TPR profiles of the Co-La-Zr/AC catalyst.

at higher reduction temperatures inhibited Pt particle agglomeration. On the other hand, we also postulate that the higher reduction temperature leads to a higher degree of cobalt reduction. When the Co-La-Zr/AC catalyst was reduced at 703 K, appropriate amounts of Co⁰ and Co₂C species were obtained. Further careful and extensive research needs to be carried out to understand the effect of Co₂C on CO hydrogenation over Co-based catalysts.

In summary, direct synthesis of higher mixed alcohols from syngas under moderate reaction conditions was achieved over a Co-La-Zr/AC catalyst reduced at 703 K. The reduction temperature affected the metal cobalt dispersion, the degree of cobalt reduction, and the amount of Co₂C species formed.

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